TASK 3: MACHINE LEARNING MODEL EXPLORATION

1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. In this activity, I explored Google Teachable Machine, a beginner-friendly platform that allows users to train and test Machine Learning models without advanced programming.

2. Objective

The objective of this task is to understand the basic working of a Machine Learning model by creating, training, and testing an image classification model.

3. Tool Used

Platform: Google Teachable Machine

Website: https://teachablemachine.withgoogle.com/

Project Type: Image Classification

Programming Requirement: No advanced coding required.

4. Project Description

I created an image classification model to identify two categories of fruits: apples and bananas. I collected example images for both categories and organized them into separate classes.

The model was trained using these images to learn visual patterns that help distinguish between the two types of fruit.

5. Procedure

Step 1: Opened Google Teachable Machine.

Step 2: Created a new Image Project.

Step 3: Created two classes named Apple and Banana.

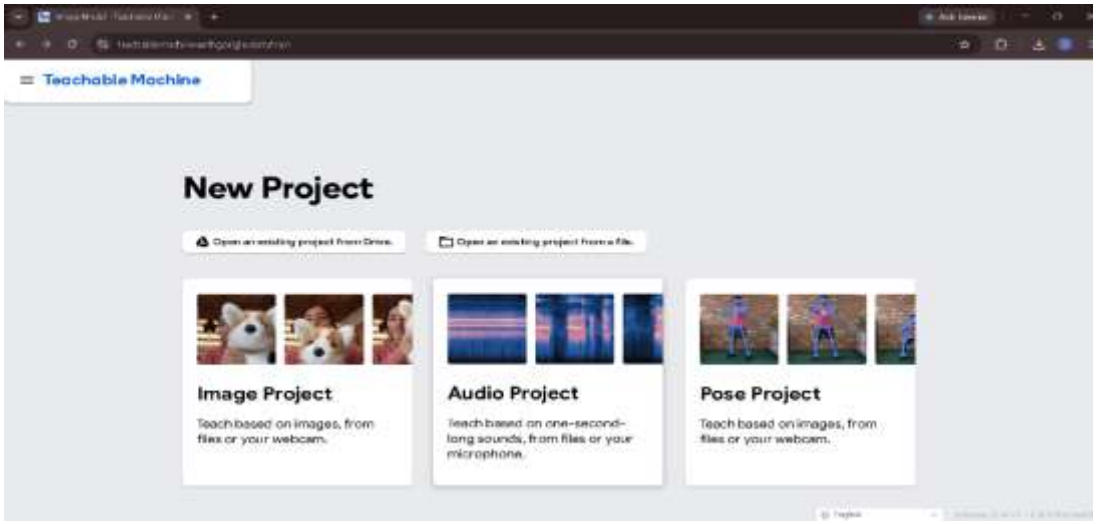
Step 4: Added example images to both classes.

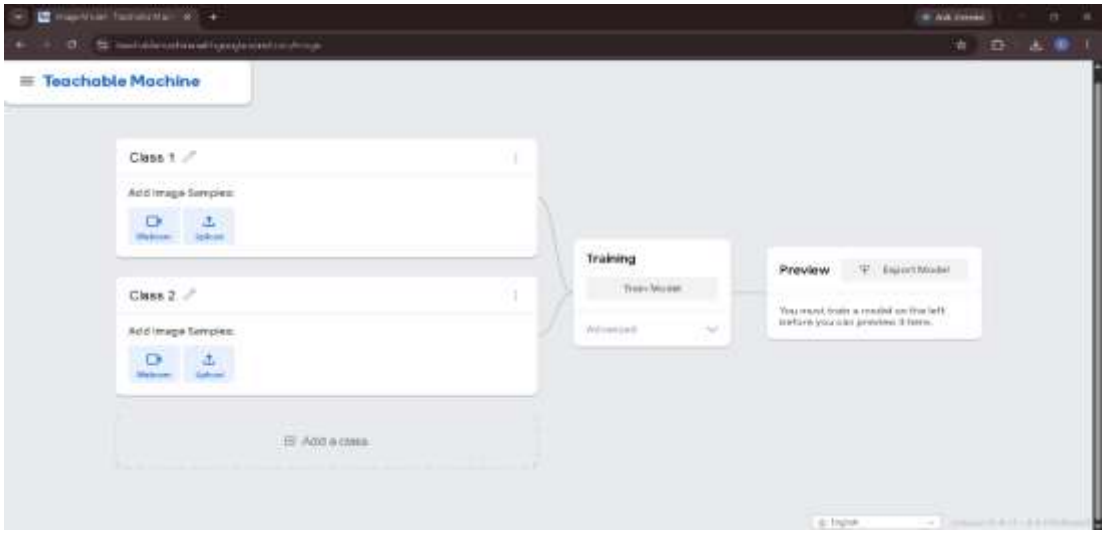
Step 5: Trained the model using the collected images.

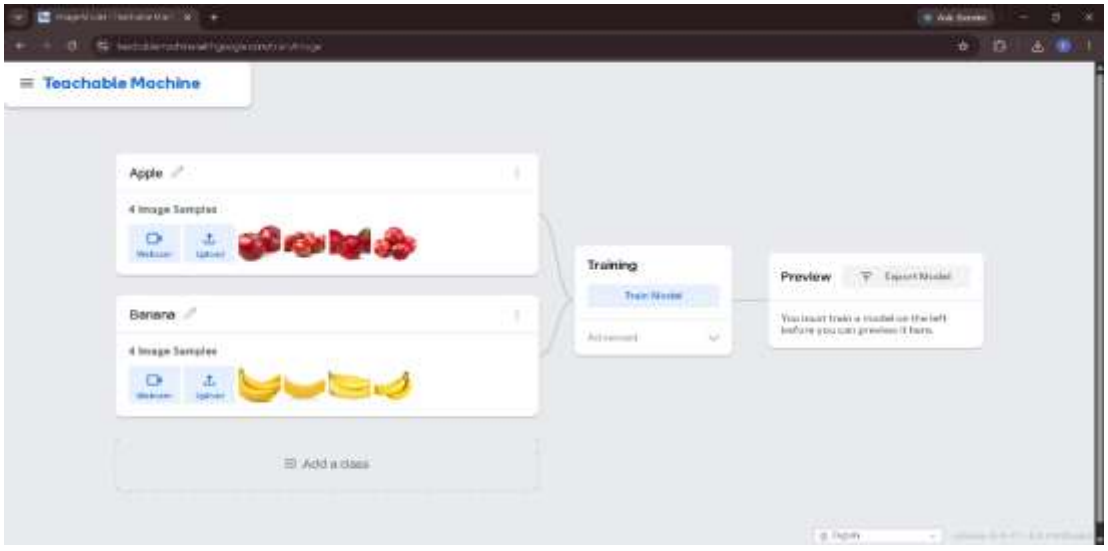
Step 6: Tested the trained model with apple and banana images.

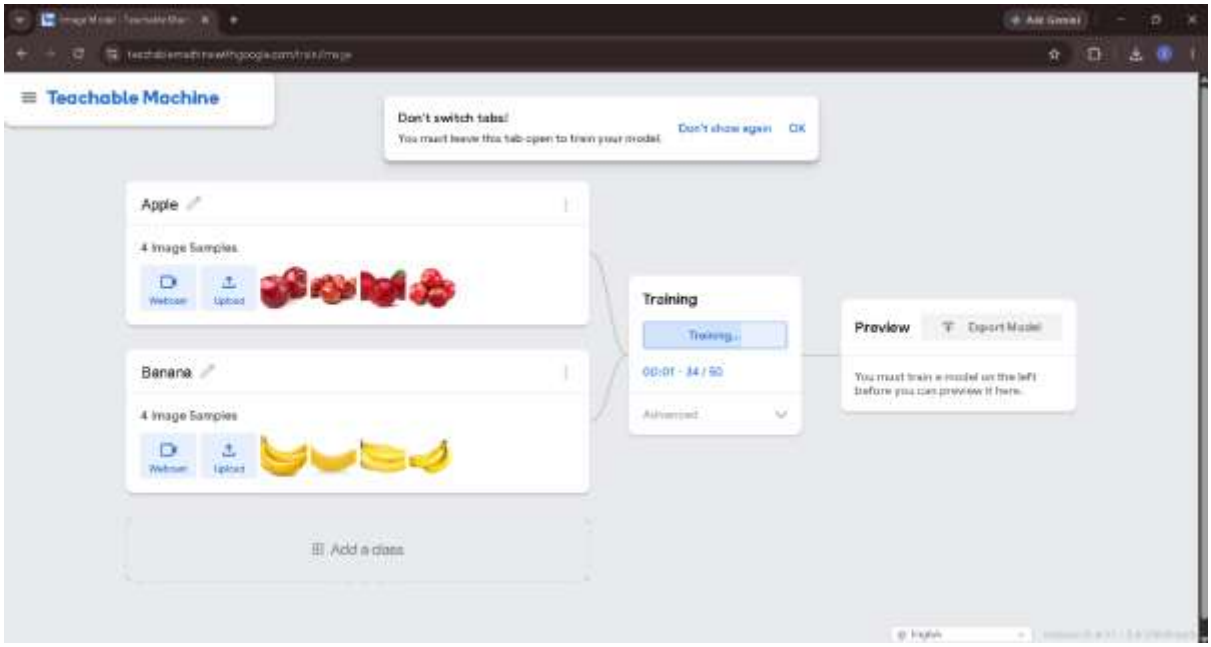
Step 7: Observed the predictions and confidence scores displayed by the model.

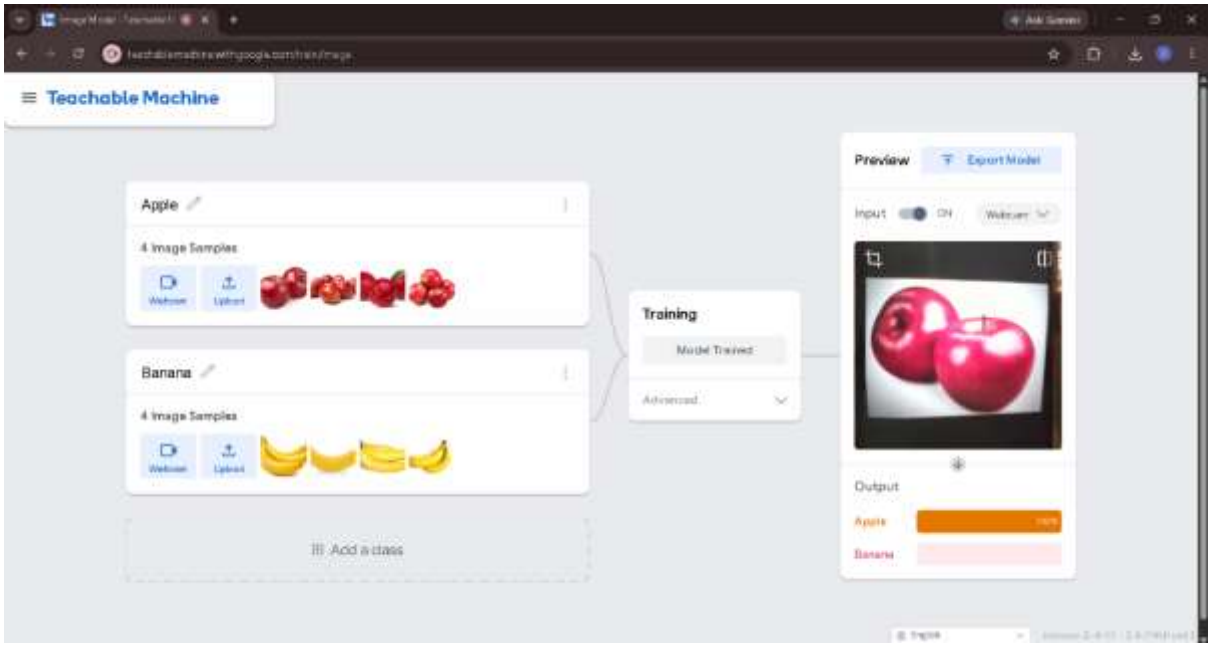
6. Screenshots and Observations


Teachable Machine
New Project
Image Project
Teach based on images, from files or your webcam.
Audio Project
Teach based on one-second-long sounds, from files or your microphone.
Pose Project
Teach based on images, from files or your webcam.


Teachable Machine
Class 1
Add Image Samples:
Class 2
Add Image Samples:
Add a class
Training
Preview


Teachable Machine
Apple
4 Image Samples
Banana
4 Image Samples
Add a class
Training
Preview



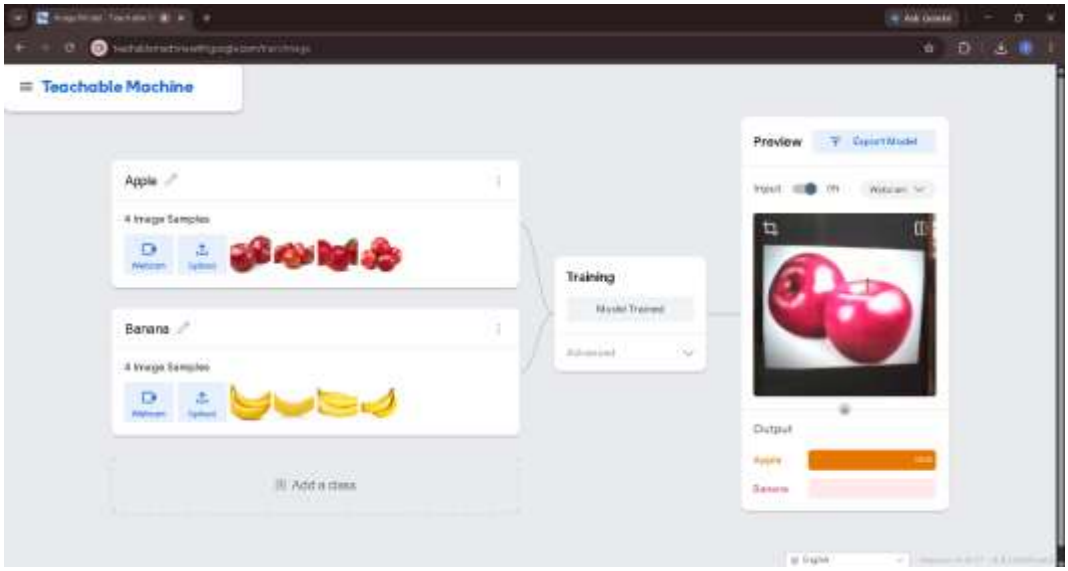


7. Results

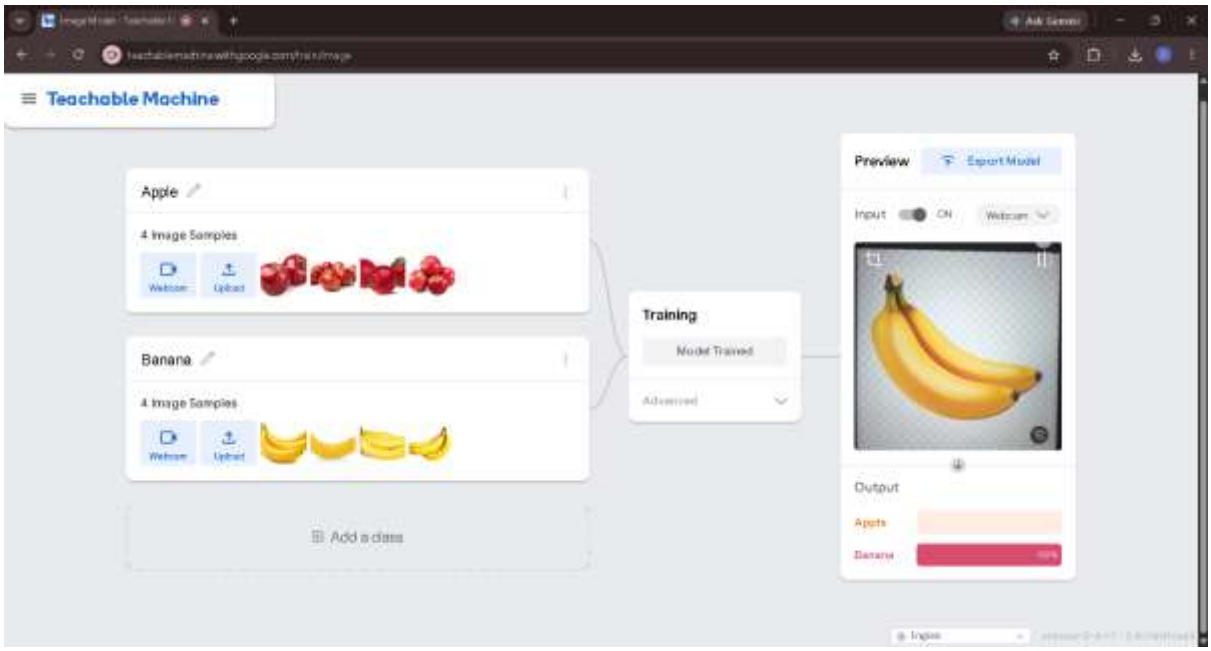
The model was trained to classify images into two categories: Apple and Banana. I tested it using example images and observed its predictions and confidence scores.

Actual testing results:

Apple prediction:



Banana prediction:



The results demonstrate how a Machine Learning model can learn from example images and use the learned patterns to classify new inputs.

8. Learning Outcomes

1. Understood the basic concept of Machine Learning.
2. Explored a beginner-friendly Machine Learning platform.
3. Learned how to create classes and collect training examples.
4. Gained practical experience in training an image classification model.
5. Learned how to test a model and interpret its predictions.

6. Understood that prediction accuracy depends on the quality and variety of training data.

9. Conclusion

This activity provided practical experience with Machine Learning using Google Teachable Machine. I learned how to create an image classification model, train it with sample images, and evaluate its predictions. The activity helped me understand the basic Machine Learning workflow without requiring advanced coding skills.